Lakshya JEE 2.0 (2025)

Maths

DPP: 4

Matrices

Q1 If $A=\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$, then $|\mathrm{A}||\operatorname{adj} \mathrm{A}|$ is equal to:

(A) a^{25} (B) a^{9}
(C) a^{81} (D) None of these

Q2 If A is a 3×3 skew-symmetric matrix, then $|A|$ is:
given by

(A) 0 (B) -1
(C) 1 (D) None of these

Q3 If $\mathrm{A}=\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$, then $\operatorname{adj}(\operatorname{adj} \mathrm{A})$ is:

(A) $\begin{bmatrix} 3 & 3 & 3 \\ 6 & 9 & 15 \\ 9 & 15 & 36 \end{bmatrix}$
(B) $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$
(C) $\begin{bmatrix} 3 & 6 & 9 \\ 3 & 9 & 15 \\ 3 & 15 & 36 \end{bmatrix}$
(D) None of these

Q4 Let A be a square matrix of order 3 such that transpose of inverse of A is A itself. Then $|\operatorname{adj}(\operatorname{adj} A)|$ is equal to:

(A) 9 (B) 27
(C) 4 (D) 1

Q5 Let $A=\begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix}$. If $\left|A^{2}\right|=25$, then $|\alpha|$ equals:

(A) 5^{2} (B) 1
(C) $1/5$ (D) 5

Q6 If $A=\begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and A adj $A=AA^{T}$, then $5a+b$ is equal to:

(A) 13 (B) -1
(C) 5 (D) 4

Q7 Let k be a positive real number and let

$$\mathrm{A}=\begin{bmatrix} 2\mathrm{k}-1 & 2\sqrt{\mathrm{k}} & 2\sqrt{\mathrm{k}} \\ 2\sqrt{\mathrm{k}} & 1 & -2\mathrm{k} \\ -2\sqrt{\mathrm{k}} & 2\mathrm{k} & -1 \end{bmatrix} \text{ and }$$

$$\mathrm{B}=\begin{bmatrix} 0 & 2\mathrm{k}-1 & \sqrt{\mathrm{k}} \\ 1-2\mathrm{k} & 0 & 2\sqrt{\mathrm{k}} \\ -\sqrt{\mathrm{k}} & -2\sqrt{\mathrm{k}} & 0 \end{bmatrix}$$

If $\det(\operatorname{adj} A)+\det(\operatorname{adj} B)=10^{6}$, then $[k]$ is equal to:
(where [.] is G.I.F.)

(A) 3 (B) 2
(C) 5 (D) 4

Q8 If $A=\begin{bmatrix} 4 & 2 \\ 3 & 4 \end{bmatrix}$, $|\operatorname{adj} A|$ is equal to

(A) 6
(B) 16
(C) 10
(D) None of these

Q9

If A and B are symmetric matrices of the same order, then:

(A) AB is a symmetric matrix

(B) $\mathrm{A}-\mathrm{B}$ is skew-symmetric matrix

(C) $\mathrm{AB}+\mathrm{BA}$ is a symmetric matrix

(D) $\mathrm{AB}-\mathrm{BA}$ is a symmetric matrix

Q10 If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=5$ then the value of $\begin{vmatrix} b_2c_3-b_3c_2 & a_3c_2-a_2c_3 & a_2b_3-a_3b_2 \\ b_3c_1-b_1c_3 & a_1c_3-a_3c_1 & a_3b_1-a_1b_3 \\ b_1c_2-b_2c_1 & a_2c_1-a_1c_2 & a_1b_2-a_2b_1 \end{vmatrix}$

(A) 5 (B) 25

(C) 125 (D) 0

Q11 If $\mathrm{A}=\begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then adj $\left(3\,\mathrm{A}^2+12\,\mathrm{A}\right)$ is equal to:

(A) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$

(B) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$

(C) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

(D) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$

Q12 If $\mathrm{A}=[a_{ij}]_{4\times4}$ such that $a_{ij}=\begin{cases} 2, & i=j \\ 0, & i\neq j \end{cases}$,, Then $\left\{\frac{1}{7}\det(\operatorname{adj}(\operatorname{adj} A))\right\}$ is, where $\{\cdot\}$ denotes the fractional part function.

(A) $1/7$ (B) $2/7$

(C) $3/7$ (D) None of these

Q13 If the matrices $A=\begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B=$ adj A and $\mathrm{C}=3\,\mathrm{A}$, then $\frac{|\operatorname{adj}\mathrm{B}|}{|\mathrm{C}|}$ is equal to

(A) 72 (B) 8

(C) 16 (D) 2

Answer Key

Q1	(B)
Q2	(A)
Q3	(C)
Q4	(D)
Q5	(C)
Q6	(C)
Q7	(D)
Q8	(C)
Q9	(C)
Q10	(B)
Q11	(C)
Q12	(A)
Q13	(B)

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